Course Code : ECT 451-3 **KQLR/RS – 24 / 1674**

Seventh Semester B. Tech. (Electronics and Communication Engineering) Examination

DIGITAL IMAGE AND VIDEO PROCESSING

Time : 3 Hours] [Max. Marks : 60

Instructions to Candidates :—

(1) All questions are compulsory and carry marks as indicated.
(2) Assume suitable data wherever necessary.
(3) Illustrate your answer with neat sketches.
(4) Use of Non – programmable calculator is permitted.

1. (a) A gray scale image on paper having physical dimensions 3 inch x 2·2 inch is scanned at 150 dpi. Calculate the following parameters :

(1) Bits require to represent the image.

(2) Time required to transmit the image if transmission speed is 56 kbps. 2(CO1)

(b) Distinguish between gray image, color image and binary image. Give suitable example to each type of images. 6(CO1)

(c) An image having a pixel values of (R, G, B) = (140, 45, 10). Calculate its value for CMY and HSI color model. 2(CO1)

2. (a) Apply DFT to the following image f(x, y) :

$$f(x, y) = \begin{bmatrix} 7 & 0 \\ 3 & 1 \end{bmatrix}$$ 3(CO3)

(b) Apply DCT to the image given in Fig. 2(b) and truncate the values below 0·04. After truncation reconstruction the image by taking IDCT.

2	0	0	0
1	2	1	1
1	0	0	1
1	1	2	1

Fig. 2 (b) 7(CO3)

3. (a) The Table 3 (a) below specifies the histogram of image I(5, 5). Perform its histogram equalization.

Gray Level	0	1	2	3	4	5	6	7
No. of Pixels	0	0	1	6	11	5	2	0

Table 3 (a) 6(CO3)

(b) Explain contrast stretching and intensity level slicing. 4(CO3)

4. (a) Explain JPEG compression method with detail block description and enlist its advantages. 6(CO3)

(b) List out BT.601 Chrominance sub-sampling formats for different digital video applications and explain any one. 4(CO2)

5. (a) Discuss in details the perspective projection model for video representation from its 3-D position. 6(CO2)

(b) Differentiate between motion representations :

(a) Pixel-based and

(b) Global. 4(CO4)

6. (a) Describe Hierarchical Feature Matching Motion Estimation Scheme (HFM-ME). 5(CO4)

(b) Develop an image processing system for orientation correction which will accept a binary image shown in Fig. Q. 6b-1 and generate the output image shown in Fig. Q. 6b-2. Draw the block diagram of the system you have proposed and explain its functioning.

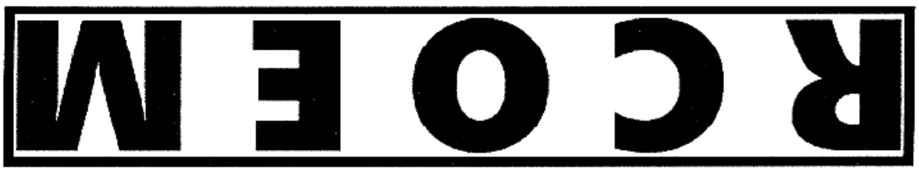

Fig.Q.6b-1

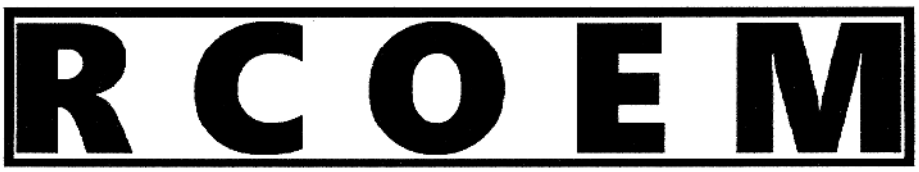

Fig.Q.6b-2 5(CO5)